

ሰራተኛ Serategna

The work-to-credit operating system for Ethiopia's real economy

Application Documentation — Phase 1: The Marketplace

Master Specification v3.0

Prepared 17 June 2026

Mo Creatives · Addis Ababa · Confidential

Contents

1. Executive overview
2. Technology stack & languages
3. System architecture
4. Data model
5. Feature catalogue
6. Matching algorithm & Serategna Score
7. Payments & regulatory compliance
8. Security & privacy
9. Business & operating model
10. API surface
11. Running the app (local & Docker)
12. Known gaps & roadmap (what's overlooked)
13. Demo accounts

1. Executive overview

Serategna (Amharic for *worker*) is a two-sided marketplace that converts verified, paid informal work into a portable financial identity. It connects households, businesses and institutions with vetted workers — from housemaids and tradespeople to formal contract staff — while eliminating the traditional broker (*delala*): no broker fee, transparent fair wages, and a guarantor-backed trust layer. Every completed job builds a worker's **Serategna Score** (300–850) and a shareable **Verified Income Passport**, creating the proprietary earnings dataset that unlocks credit in later phases.

What makes it distinctive

- **Housemaid-first:** a front-and-centre flow to hire a trusted permanent maid — mandatory legal guarantor (ዋስ / *wastina*), enforced minimum wage, flat ETB 100/month, up to 4 posts/year.
- **Direct pay, never held:** the platform never custodies money. Employers pay workers directly (Telebirr / CBE Birr / cash); workers keep 100%. Funded by a flat employer subscription.
- **80% skill-match gate:** workers only see jobs (and clients only see workers) at $\geq 80\%$ relevance, in both directions, with transparent match reasons.
- **Matchmaking UX:** Uber/dating-app inspired swipe decks for both job discovery and worker shortlisting, with live taxi-style tracking and ETA during an engagement.
- **Trust infrastructure:** tiered Fayda identity, verifiable institutional certifications, a psychometric reliability assessment, two-way ratings, reputation badges and an SoS safety desk.
- **Compliant by design:** no stored value and no held funds means no NBE Payment Instrument Issuer licence is required.

2. Technology stack & languages

The entire system is built on the TypeScript / Node ecosystem. **There is no Python.**

Language	Where it is used	Footprint
TypeScript	Web frontend and backend API (end-to-end)	56 .ts + 38 .tsx files
JavaScript	Build/config & scripts (Tailwind, Postgres switch)	~3 files
SQL	Generated Postgres schema (docs/db)	2 .sql files
Prisma schema (DSL)	Canonical data model	1 schema file
HTML / CSS	App shell + Tailwind utility CSS	1 .html + index.css
Shell / Batch / YAML	START.bat, setup scripts, docker-compose	several

Frameworks & runtimes

Tier	Stack
Web app (the live product)	React 18 · Vite · React Router · Tailwind CSS · TypeScript
Backend API	Node.js · Express · Prisma ORM · Zod · JWT · Helmet · TypeScript (tsx)
Database	SQLite in development · PostgreSQL in production (via Prisma)
Mobile	Expo / React Native — present as a scaffold only (not yet built out)
Infrastructure	Docker · docker-compose (Postgres + API + web)

Note on mobile: the `mobile/` folder contains an Expo + React Native starter, but its screens are the default template — none of the Serategna features are implemented there. The fully-built product is the web app, which is mobile-first and installable from a browser.

3. System architecture

Serategna is a monorepo with three apps and a shared docs set.

```
serategna-v3/  
■■■ server/ Express API + Prisma (jobs, score, ledger, safety, admin)  
■■■ web/ React + Vite app (worker + client + admin in one bundle)  
■■■ mobile/ Expo / React Native scaffold  
■■■ docs/ architecture, security, business, algorithm, SQL schema, spec PDFs  
■■■ docker-compose.yml
```

Key design decisions

- **One web bundle, three audiences:** a role switch flips between Worker and Client modes; admins reach a separate operations console. Role-based accent colours (cyan for workers, indigo for employers) keep the two sides visually distinct.
- **Append-only double-entry ledger:** retained for reconciliation integrity even though funds are no longer held — entries are now notional direct employer→worker records that always reconcile to zero.
- **Adapter seams for integrations:** payments, SMS, push (FCM), Telegram and Fayda/MOSIP identity all sit behind swappable adapters with a mock/fallback default and an admin status panel.
- **Server-side enforcement:** wage floors, the 80% match gate, posting/application quotas and phone-number masking are all enforced in the API, not just the UI.
- **Dev simplicity:** SQLite by default means zero database to install locally; a script flips Prisma to Postgres for Docker/production.

4. Data model

The Prisma schema defines **29 models**. The principal entities:

Domain	Models
Identity & access	User, RefreshToken, OtpCode, VerificationRequest, AuditLog, Consent
Profiles	WorkerProfile, BusinessProfile, Agent
Work & matching	Job, Bid, Message, SavedItem, PriceBand
Money (notional)	LedgerEntry, Payout
Trust & reputation	Rating, Certification, PsychometricResult, ScoreSnapshot
Agreements	Contract, ContractSignature, Guarantor
Safety & ops	Dispute, SosEvent
Growth & comms	Subscription, Lead, Notification, NotificationLog

Job records carry the full engagement lifecycle (open → accepted → enroute → started → completed → paid → confirmed), scheduling (startsAt / endsAt), an application deadline (expiresAt, auto-expired), live GPS fields for tracking, and domestic-contract details (live-in, duties, day-off, guarantor-required).

5. Feature catalogue

Discovery & matching

- Swipe decks both ways — workers quick-bid jobs; clients shortlist workers (persisted).
- Transparent 80% skill-match gate with per-match reasons; one relevance algorithm ranks both sides.
- Google Maps pin-drop location, worker-on-map browse, list/map and swipe/list toggles (styled fallback without an API key).
- Job application deadlines with a live 'closes in' countdown and automatic removal when expired.

Work types & posting

- Gigs, short-run/daily, fixed-term contract, permanent placement, and group hire — formal & informal.
- Minimum / living wage enforced on every job, on both posting and bidding.
- AI-assisted job drafting and AI CV builder/import (currently rule-based parsers, see §12).
- Trilingual job taxonomy — 20 groups, 270+ roles in English, Amharic and Afaan Oromoo.

Trust, safety & identity

- Tiered Fayda identity verification; verifiable institutional certifications (admin-reviewed).
- Psychometric reliability assessment feeding the Score and matching.
- Guarantor (የስራ ዋስ / wastina) suretyship; digital consensus + printable in-person legal contract.
- Two-way ratings, reputation badges, SoS safety desk, phone-number masking, female-only client option.

Identity, growth & platform

- Shareable Verified Income Passport (public profile + QR, no login required).
- Employer subscription (ETB 100/mo or 1000/yr, 5 posts/mo); worker 5 applications/month.
- Enterprises packages (Team, Institution, Serategna-Managed) with lead capture and a call centre.
- Standalone marketing landing page, light/dark themes, toasts, skeletons, in-app notifications.
- Admin console: verifications, certifications, disputes, SoS, leads, reconciliation, KPIs, integrations, and a DPO data-subject access tool.

6. Matching algorithm & Serategna Score

Match score (deterministic, skill-first)

Relevance is computed from a skill-fit base plus a weighted quality term, then scaled and gated at 80%:

```
fit = roleMatch ? 75 : categoryMatch ? 68 : 12
quality = (proximity·0.35 + score·0.30 + rating·0.20 + verified·0.10 + freshness·0.05) · 25
total = clamp(fit + quality) · ×0.6 if the worker is not currently available
```

Serategna Score (300–850)

Component	Weight	Drivers
Transaction integrity	35%	Completed jobs, confirmations, no-shows/disputes
Earnings consistency	30%	Verified income regularity over time
Platform behaviour	20%	Conduct + psychometric reliability index
Relationship capital	15%	Ratings, repeat hires + verified-certification boost

Every match surfaces *why* it ranked (skill, proximity, score, rating, verification, freshness), so neither side sees an opaque black box.

7. Payments & regulatory compliance

- **No funds custody.** Serategna never holds, stores or transfers user money. Payment happens directly between employer and worker off-platform (Telebirr, CBE Birr or cash).
- **No NBE PI licence required.** Because there is no stored value or held funds, the platform avoids the National Bank of Ethiopia Payment Instrument Issuer licensing trigger.
- **Revenue = subscription, not commission.** Workers keep 100% of wages; the platform is funded by the flat employer subscription. Ledger entries are notional and reconcile to variance zero.
- **Legal framework.** Digital consensus records in-app; the binding contract is signed in person (Proclamation 1205/2020). Guarantor suretyship follows 1243/2021. Data handling follows Personal Data Protection Proclamation 1321/2024, with a consent ledger and an audited DPO access tool.

8. Security & privacy

- OTP authentication with brute-force lockout (5 attempts), per-phone cooldown and per-IP rate limiting; JWT access + rotating refresh tokens.
- Hardened HTTP — Helmet with a strict Content-Security-Policy, HSTS preload, x-powered-by disabled.
- Tamper-evident AuditLog (with IP) on sensitive admin actions; phone numbers masked unless owned, admin, or an engaged counterparty on an active job.
- No funds custody removes the single largest class of financial risk.
- DPO data-subject access is restricted, logged, and framed under Proclamation 1321/2024.

9. Business & operating model

Stream	Model
Employer subscription	ETB 100 / month or 1000 / year — 5 job posts per month
Permanent housemaid	Capped at 4 posts / year; guarantor mandatory
Worker activity	5 job applications per month (free)
Enterprises — Team	ETB 1,500 / month · 25 posts · 2 manager seats
Enterprises — Institution	ETB 6,000 / month · 100 posts · 5 seats · private talent pool
Enterprises — Managed	Custom by agreement — Serategna recruits, vets & rosters

Phase 1 builds the verified-earnings dataset. Later phases monetise that asset through credit and financial-identity products — the strategic reason the Score and income passport are built from job one.

10. API surface

The API exposes 23 route groups under /api:

```
auth, profiles, jobs, bids, catalog, score, ratings, wallet, disputes, sos, identity,
contracts, guarantors, credentials, legal, notifications, saved, subscription, enterprise,
public, ai, integrations, admin
```

Highlights: **public** serves the no-login income passport; **enterprise** handles packages, contact and lead capture; **integrations** reports adapter status; **admin** covers the full operations console including the audited DPO data-subject endpoint.

11. Running the app (local & Docker)

Local development (no external services: needs Node 20+)

```
# from serategna-v3/ npm run first-run # install, set up SQLite DB, start API + Web #  
Windows: double-click START.bat # then open http://localhost:5173
```

Phone testing over LAN

The Vite dev server is exposed on the network (host: true). Scan the generated QR (Serategna_app_qr.png) with your phone camera while on the same Wi-Fi to open the app in mobile Safari/Chrome.

Docker (Postgres + API + web)

```
cp .env.example .env # set strong JWT secrets + DB password  
docker compose up --build # web on :8080, API proxied at /api  
# production: SEED_ON_START=false OTP_DEV_MODE=false
```

12. Known gaps & roadmap — what's overlooked

An honest assessment of what is not yet production-grade, by priority.

Critical (block a real launch)

- **No automated tests.** Zero unit/integration/e2e tests across server and web — risky for a money-adjacent, safety-critical app.
- **'AI' features are rule-based, not AI.** The CV builder, job-draft and license import use deterministic text parsers, not a real LLM (e.g. Claude). They are branded as AI but do not call a model.
- **Integrations are stubs/fallbacks.** Payments (Chapa), SMS, FCM push, Telegram and Fayda/MOSIP all default to mock. OTP is written to the server console, not actually delivered by SMS.
- **No payment proof.** Since pay is off-platform, there is no receipt upload or transaction-reference capture to back disputes — payment confirmation is trust-based.
- **Feed does not scale.** The job feed loads all open jobs into memory and filters/sorts in JavaScript; there is no DB-level geo query, indexing strategy or pagination.

High value (needed soon)

- **Real-time is polling.** Chat, notifications and live tracking poll on a timer; no WebSocket/SSE, so updates lag and waste battery/data.
- **Low-literacy onboarding.** The target demographic (housemaids) may have limited literacy — no voice/IVR/USSD, audio prompts or agent-assisted signup yet, and most UI strings are English-only despite the trilingual taxonomy.
- **Offline / PWA.** No service worker or offline shell; informal-economy users often have intermittent connectivity.
- **Dispute evidence.** Disputes capture a text reason only — no photos, timestamps or structured evidence.
- **Observability.** No error tracking (e.g. Sentry), structured metrics, or health/readiness probes for production.

Medium (hardening & trust)

- DB migrations use prisma db push (no migration history) and there is no documented backup/restore strategy.
- Account recovery (lost SIM / number change) and admin 2FA / fine-grained roles are not implemented.
- Anti-abuse beyond OTP throttling — fake-account / fake-review detection, device fingerprinting, review moderation.
- Cancellation, no-show and partial-completion policies, and worker background-check integration.
- Guarantor identity is stored but not independently verified; subscription VAT/TOT receipts (ERCA) are not generated.
- Accessibility pass (screen-reader labels, focus order, contrast) and CI/CD pipeline beyond the placeholder.

None of these are blockers for a pilot or investor demo — the end-to-end product flows work and reconcile. They are the gap between a strong prototype and a hardened, at-scale production service.

13. Demo accounts

In development the OTP is auto-filled. Sign in with any of:

Role	Phone
Client (Janani)	+251922000001
Worker · Tier 1 (Hanna)	+251911000001
Worker · Tier 0 (Meron)	+251911000007
Admin / Operations	+251900000000

Serategna · ስራተኛ — Work · Build credit · Thrive. Prepared for Mo Creatives, Addis Ababa. Confidential.